Rithish Murugan **Software Engineer**

United States | +1 (773) 930-2475 | muruganrithish4@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

Results-driven Software Engineer with 3+ years of experience building intelligent systems that solve real business problems from automating support operations and eliminating manual workflows to detecting financial fraud at scale. Delivered measurable impact across startups, SaaS platforms, and Fortune 500 banking institutions. Recognized for translating complex AI and backend engineering challenges into production-ready systems that teams rely on daily and businesses measure in dollars saved and hours recovered. Passionate about building at the intersection of AI and real-world impact, currently seeking a mid-level Software engineering role where intelligent systems meet meaningful problems.

EXPERIENCE

eAlliance Corporation, Software Engineer — Jul 2025 – Present | Remote, USA

- Design an end-to-end AI automation platform using **Python, FastAPI, and AWS Bedrock,** integrating **LLMs, RAG, and multi-agent orchestration,** reducing operational support time by 30% across 500+ daily requests.
- Engineer a production **RAG pipeline** using **AWS Bedrock and Pinecone vector database** for document ingestion, semantic search, and **NLP-based classification**, processing thousands of documents for internal chatbot automation.
- Develop an autonomous AI agent with multi-step reasoning, prompt engineering, and RBAC using **FastAPI,** automating vendor-matching workflows and eliminating manual decision-making across cross-functional engineering and operations teams.
- Architect scalable **RESTful APIs** on **AWS Lambda and S3** using **serverless and microservices** best practices, reducing internal service latency by 25% across distributed AI inference pipelines serving multiple internal teams.
- Build real-time internal dashboards using **JavaScript and Supabase** with live API integration, surfacing actionable operational insights and reducing manual lookup time by 50% across cross-functional business and engineering teams.
- Maintain production backend services across staging and prod environments, conducting model performance evaluation, applying **CI/CD practices**, and automating **Python and Supabase workflows** to reduce system failures by 20%.

Virtualan Software LLC, Software Engineer — May 2024 – Aug 2024 | Chicago, IL

- Automated scalable backend data pipelines using **Python and SQL,** applying ETL best practices and workflow automation to reduce data processing time by 40% across 10,000+ daily records in critical reporting systems.
- Designed, developed, and validated **RESTful APIs** using **Postman and Swagger**, improving API test coverage by 35%, system reliability, and backend service quality across cross-functional engineering and QA teams.
- Integrated **AWS S3 and Redshift** with backend workflows, implementing cloud-based data storage and query optimization strategies to improve data scalability and accelerate query performance by 25% across production pipelines.
- Collaborated with cross-functional QA engineers to implement role-based **access controls and validate data pipelines,** applying security best practices and reducing production bugs by 20% across backend systems.
- Supported backend architecture discussions and code reviews during **Agile sprint planning,** driving engineering best practices and improving team delivery velocity by contributing to 3 feature releases across iterative development cycles.

Hexaware Technologies, Software Developer — Mar 2021 – Jul 2023 | Chennai, India

- Engineered 10+ production-grade **Java Spring Boot microservices** for a Fortune 500 US banking client, securing 1M+ daily transactions through **OAuth2, JWT** tokenization, rate limiting, and fault-tolerant **API architecture** across distributed systems.
- Collaborated in building an **ML-powered risk scoring engine** integrating **OpenAI APIs, LangChain**, and **Hugging Face Transformers**, achieving 92% fraud detection accuracy and reducing false positive alerts by 28% saving $2M+ annually.
- Architected a high-throughput **Apache Kafka** event streaming pipeline processing 500K+ transactions per hour, leveraging event-driven microservices and asynchronous processing to detect fraud with sub-second latency.
- Accelerated release velocity by 40% by containerizing microservices with **Docker**, orchestrating deployments on **AWS EKS** via **Kubernetes**, and establishing **Jenkins CI/CD pipelines** sustaining 99.9% uptime for a 12-member cross-functional Agile team.
- Boosted platform performance by 50% through **PostgreSQL** and **MongoDB** query optimization and **Redis caching**, reducing database load by 45% while delivering a **React.js** real-time fraud analyst dashboard with 80%+ test coverage.

SKILLS

Programming Languages: Python, R, Java, JavaScript, SQL, C, C++, HTML, CSS
Frameworks & Libraries: FastAPI, Spring Boot, React.js, LangChain, Hugging Face Transformers
AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, Prompt Engineering, NLP, Semantic Search, Fraud Detection ML, Risk Scoring Engines, Hugging Face Transformers, LangChain, OpenAI APIs, Scikit-learn, PyTorch, Model Performance Evaluation
Cloud & Infrastructure: AWS (Bedrock, Lambda, S3, EKS, Redshift), Kubernetes, Docker, Serverless Computing, Microservices Architecture, Distributed Systems
Databases & Data Engineering: Pinecone (Vector Database), Supabase, MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, ETL Pipelines, Query Optimization, Event-Driven Architecture
APIs & Backend Development: RESTful APIs, FastAPI, Spring Boot, OAuth2, JWT, Rate Limiting, Fault-Tolerant Architecture, Postman, Swagger, RBAC, Role-Based Access Control, n8n automation
DevOps & CI/CD: Jenkins, Docker, Kubernetes, AWS EKS, CI/CD Pipelines, GitHub, Agile, Sprint Planning, Code Reviews
Data Analytics: Real-Time Dashboards, Apache Kafka, Redshift, ETL, Data Pipelines, Workflow Automation, Business Intelligence

EDUCATION

Master of Science in Computer Science — Aug 2023 – May 2025 | Chicago, IL
Illinois Institute of Technology